

THE MATH OF TIME

$$P (1 + r/n)^{nt} = A$$

P = principal, the amount you start with

r = the interest rate, written as a decimal

n = how many times per year the interest gets added

t = time in years

A = the amount you end up with

FROM THE BOOK'S OWN EXAMPLE

\$100 at 8%, added once a year

\$136.05 after 4 years \$466.10 after 20 years

Added TWICE a year instead: \$480.10

Same money. Same rate. Same 20 years.

One formula. Four situations.

SAVINGS

\$1,000 at 4%, added monthly,
left alone 5 years

\$1,221.00

Slow, steady, yours.

INVESTING

\$2,000 at 8%, added yearly,
left alone 30 years

\$20,125.31

Time does the heavy lifting.

CREDIT CARD

a \$1,200 balance at 24%, added
monthly, untouched 3 years

\$2,447.86

It more than doubles. Against you.

CAR LOAN

borrow \$20,000 at 7% -- first
month's interest alone:

\$116.67

Taken monthly on what is left.

The formula doesn't care whose side it's on. Put it on yours.